Practice Exam - 1

Question 40 of 117

A 72-year-old patient with end-stage heart failure NYHA class IV with non-ischemic cardiomyopathy was referred to you for evaluation for LVAD therapy. His ECG showed sequential AV pacing at 70/min with BP of 100/54 mmHg. His blood tests showed creatinine 1.3mg/dl, sodium 132 mEq/l and pro-BNP 5,455 pg/ml.

Echo doppler revealed markedly dilated LV cavity with LVEF 15%, moderate MR and moderate TR with mildly enlarged RV chamber. Doppler showed an increased E/E' ratio of 20. Right heart catheterization showed RA pressure of 22mmHg, PCWP of 24mmHg, mean PA of 60/30/45mmHg, cardiac index of 1.8 l/min/m2, and PVR of 4.4 Wood Units.

Which of the following hemodynamic and echnocardiographic measures are most predictive of RV failure after LVAD implantation

A. Severe pulmonary hypertension with enlarged right ventricle

B. RV stroke work index of 658 mmHg-ml/m2 and moderate TR

C. Moderate MR and moderate TR with high E/e' ratio of 20

D. RA pressure/PCWP ratio of 0.9 and PAPi (pulmonary artery pulsatility index) of 1.36

E. TAPSE (tricupsid annular plan systolic excursion) of 1.6cm and high E/E' ratio of 20

Commentary | References

Testing Point: The AHFTC specialist should know which data are most useful for predication of RV failure after LVAD implant Question Line: Which of the following combinations of hemodynamic measures or echo doppler findings are the most predictive of RV failure after LVAD implantation? Distractors: valve regurgitations, severe pulmonary HTN, high LV filling pressure.

Answer: D. RA pressure / PCWP ratio of 0.9 and PAPi (pulmonary artery pulsatility index) of 1.36.

RV failure after LVAD placement is a frequent complication, and therefore, assessment of RV function is very important prior to LVAD implant. Post-LVAD RV failure decreases success of LVAD therapy and increases morbidity and mortality in these patients. A high CVP/wedge ratio is a known predictor of poor RV function. In clinical trials and registries, PAPi, RVSWI and CVP/PCWP ratio were superior to other variables in predicting RV failure and need for RV assist device support following LVAD implantation. PAPi is the pulmonary artery pulsatility index and is calculated: (systolic PA – diastolic PA) / RAP. Based on clinical trials, PAPi >2 seems to be protective for RV when considering LVAD. A calculated PAPi for this case is 1.36. There is also a different cut off value for RV Impella placement in cardiogenic shock; this is <0.9. Low PAPi is associated with elevated RAP, elevated mean PAP, elevated wedge and lower cardiac index. PAPi is prognostic of CV death and HF readmissions at 6 months. CVP/PCWP ratio is a good marker of RV dysfunction. CVP>20mmHg and CVP/PCWP ratio >0.6 correlate with poor RV function.

Answer A: Although an enlarged right ventricle is of concern prior to LVAD placement, no information is provided here about RV function, which is the most important parameter. The presence of severe pulmonary hypertension suggests a compensated RV able to generate these pressures. Since LVAD therapy is highly effective at reducing PA pressure, this is often a reason to proceed.

Answer B: RVSWI is RV stroke work index and is calculated: [(cardiac index/HR) x (mean PAP-mean RAP)]. A value of <600 mmHg*mL/m2 is a predictor of risk of prolonged RV support post LVAD. In this example, RV stroke work index is above that. Recent data suggest that PAPi is a better predictor than RVSWI.

Answer C and E: High E/E' ratio correlates with high LV filling pressures and is expected to improve markedly after VAD therapy.

Answer C: Mitral or tricuspid valve insufficiency are frequent findings in advanced heart failure and may change with volume status. This information does not directly provide information about degree of RV dysfunction. These valvular findings is not a contraindication to LVAD.

Answer E: TAPSE may not be accurately measured and the TAPSE of 1.6cm in answer E is consistent with mild RV dysfunction. Know well RV function prior to final decision re: LVAD implantation. Know how to calculate RVSWI and PAPi.